COURSE OUTCOME: 1

ASSESSMENT TOOL: 1

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1. Library Management System

a) Object - Oriented Concepts

class - A class is a blueprint used to create objects. In a library management System, classes can be Book, User, Librarian & Transaction

object - An object is an instance of class. Eg: Book ("Java", "James Gosling") is an object of the Book class.

Encapsulation Data & methods are combined into a single unit. Private data such as bookID & availability can only be accessed through getter & setter methods.

Inheritance One class inherits the properties from another.

Example: student & Faculty inherit from users class.

b) object Relationships

Association : A user borrows a Book
Aggregation : A Library contains many books, but books can exist independently.
Composition : A Book consist of Book Details, if book is deleted, its details also deleted.

c) benifits

These concepts improve code reuse, modularity, reduce code duplication.

2. Consider developing a Student Information System

a) Object Oriented Design Principles

Abstraction : Abstraction hides unnecessary details.

Encapsulation: Encapsulation protects data

Modularity : Modularity divides the system into smaller parts.

Eg : Attendence module works independently.

b) Single Responsibility + Open/closed Principle.

Single Responsibility means one class performs one task,

Open/close principle allows adding new features without changing existing code.

Eg: Adding a new grading system

c) These principles improves software quality by making system easier to maintain and reuse

3) Suppose a company plans to develop Online shopping System.

a) SDLC includes requirement Analysis, Design implementation and Maintanance. For Example testing ensures the payment page works correctly.

b) Each SDLC phase helps build software in an organised way and reduces development errors.

c) SDLC delivers reliable software by ensuring quality & reducing project risks.

4. Consider a project where requirments are clearly defined from begining.

a) SDLC models
Waterfall follows a fixed sequence, while agile develops software in small iterations. Spiral focuses on risk & iteration repeats development cycles.

b) Waterfall is suitable for fixed requirements, whereas agile is better when requirements change frequently.

c) Agile is best for dynamic projects because customer feedback can be included after every iterations.

5. Suppose a system analyst is designing an ATM system.

a) UML is a visual language used to design and understand software systems.
Eg: ATM use case diagram shows customer transactions.

b) Use case shows user actions, class shows system structure, Sequence shows interaction & Activity shows workflows.

c) UML improves communication among developers & helps create better software design.

6. Hospital Management System.

Structured design divides system into smaller modules.
Eg: Patient, Billing are separate modules.

Structured design focuses on function, while object-oriented design focuses on objects like patient & Doctors

Modular design makes large-scale systems easier to test, maintain and upgrade.